



Industrial Internship Report on Python Quiz Game

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Executive Summary

This report presents the work completed during the industrial internship in Python. The project undertaken was a command-line Python Quiz Game developed to demonstrate fundamental programming concepts through an interactive question-and-answer application.

The project allows a user to view multiple-choice questions, enter an answer, receive immediate correctness feedback, and obtain a final score and percentage after completing the quiz. The implementation uses Python data structures, loops, conditional statements, functions/input handling, string processing and basic arithmetic.

The project was developed and tested using Jupyter Notebook and a Python source file. The final project files were maintained in the GitHub repository associated with the internship. The completed quiz was tested with five questions and produced a score of 5/5, equivalent to 100%.



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1 Preface

The industrial internship provided an opportunity to connect classroom knowledge of Python programming with a practical software project. During the internship, I followed the prescribed learning resources, practiced Python concepts, completed assessments, and applied the concepts to develop a functional Quiz Game.

The selected project, Python Quiz Game, was chosen because it provides a simple but practical way to demonstrate programming fundamentals. Although the application is small in scale, it involves several important programming tasks: storing structured questions, presenting options, accepting user input, comparing answers, updating a score, and displaying a final result.

The internship also introduced me to practical development tools and workflow practices, including Jupyter Notebook, Python source files and GitHub-based project submission. Working through coding errors and testing the final output helped strengthen my problem-solving approach.

I am thankful to the internship organizers, trainers and everyone involved in providing the learning resources and project framework. The internship helped me gain confidence in writing Python programs and understanding how a small software solution can be planned, implemented, tested and documented.



2 Introduction

Python is a high-level, general-purpose programming language widely used for automation, application development, data analysis, artificial intelligence and education. Its readable syntax makes it suitable for learning programming fundamentals while still being powerful enough for professional applications.

The internship focused on building practical familiarity with Python through structured learning and a project-based approach. The project component required the application of programming concepts rather than only theoretical study.

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is presented in the internship material as an organization working in the digital transformation domain, with a focus on industrial solutions, sustainability and return on investment. Its technology areas include Internet of Things, cloud computing, cyber security, machine learning, communication technologies, Java, Python and front-end technologies.

The industrial orientation of the program helped provide context for understanding how software and technology concepts can be applied to practical problems. The Python project used in this report represents the programming-focused component of that learning experience.



2.1 About UniConverge Technologies Pvt Ltd — Technology Context

UCT's technology ecosystem described in the internship material includes IoT platforms and industrial digital-transformation solutions. These solutions demonstrate how software, communication technologies, cloud services and data processing can work together to solve operational problems.

The internship material describes UCT Insight as an IoT platform designed for rapid deployment of IoT applications and business/process insights. It supports standard IoT communication protocols and can be used in cloud or on-premises environments.

The broader technology exposure was useful for understanding that programming languages such as Python are not isolated skills. They can form part of larger systems involving data processing, automation, APIs, cloud services and intelligent applications.

For my project, the main focus remained Python programming. The industrial context nevertheless helped me understand the importance of writing clear, testable and maintainable code and documenting the final solution.



2.1 About UniConverge Technologies Pvt Ltd — Solutions

The internship material also describes a smart-factory platform focused on production and asset monitoring, OEE and predictive maintenance. Such systems depend on collecting information, processing it and presenting useful results to users.

Other solution areas mentioned in the internship material include LoRaWAN-based applications for agriculture, smart cities, industrial monitoring and utility metering, as well as predictive-maintenance solutions using embedded systems, Industrial IoT and machine learning.

These examples provided useful background on how technology projects are designed around a problem, a set of inputs, processing logic and an output that provides value to a user. The same basic thinking can be applied to a smaller Python application such as the Quiz Game.

The Quiz Game therefore follows a simple input–processing–output model: questions and options form the input data, Python logic processes the user's responses, and the score and performance message form the output.



2.1 About UniConverge Technologies Pvt Ltd — Industrial Relevance

A practical software project should not be evaluated only on whether it runs once. It should also be checked for correctness, clarity, usability and predictable behaviour for different inputs. These principles were considered while testing the Quiz Game.

The project demonstrates a small example of interactive software. It receives user input, validates the response against the expected answer and maintains a running score. The final output summarizes the result for the user.

The internship's industrial orientation encouraged attention to the complete development cycle: understanding requirements, implementing functionality, running tests, identifying mistakes and preparing documentation for submission.

This experience provided a foundation that can be extended to larger Python projects involving files, databases, graphical interfaces, web applications or data analytics.



2.2 About upskill Campus

upskill Campus is part of the internship ecosystem through which the learning activities, project guidance and submission process were facilitated. The program provided a structured pathway for learning Python and completing a project within the internship period.

The learning environment combined instructional resources, assessments and practical project work. This approach helped connect individual programming concepts with an application that could be demonstrated and submitted.

The internship workflow also introduced project documentation and GitHub submission. Maintaining the project files in a repository provides a central place for reviewing the source code and notebook used for the project.

Overall, the internship structure supported progressive learning: first understanding concepts, then practicing them, and finally applying them to the selected project.



2.3 The IoT Academy

The IoT Academy is described in the internship material as the EdTech division associated with UCT and involved in delivering executive certification and technology-oriented learning programs in collaboration with academic and industry partners.

For this internship, the learning resources provided through the program helped build a foundation in Python. The practical project served as an opportunity to apply those concepts in a self-contained application.

The project-based learning model was especially useful because it required more than memorizing syntax. It required translating a simple requirement into program logic, organizing data, handling user input and validating the final behaviour.

The experience reinforced the importance of continuous practice in programming. Small projects can be used to test understanding of fundamental concepts before moving to larger applications.



2.4 Objectives of this Internship Program

2.5 Reference

2.6 Glossary

2.4 Objectives of this Internship Program

The main objectives of the internship were to gain practical programming experience, understand how learned concepts can be applied to a project, improve problem-solving skills, become familiar with development tools, and build confidence in documenting and presenting technical work.

2.5 Reference

- [1] Python documentation and standard Python programming concepts.
- [2] Learning resources and project instructions provided through the internship platform.
- [3] GitHub repository maintained for the internship project.

2.6 Glossary

Python — A high-level general-purpose programming language.

Jupyter Notebook — An interactive environment for writing and executing code in cells.

GitHub — A platform for hosting and collaborating on software repositories.

Variable — A named reference used to store a value in a program.

List — An ordered Python collection that can contain multiple values.

Dictionary — A Python data structure storing key-value pairs.

Conditional statement — A construct used to execute code based on a condition.

Loop — A construct used to repeat a block of code.

Score — The number of correctly answered questions in the quiz.



3 Problem Statement

The problem addressed by the project is to develop a simple interactive quiz application in Python that can present a set of multiple-choice questions, accept a user's answer for each question, determine whether the response is correct, maintain a score, and display a final performance message.

The application should be easy to run and understand. It should clearly display the question and four answer choices, request an answer in the expected format, provide immediate feedback, and continue until all questions have been attempted.

Functional requirements

1. Display a welcome message.
2. Store multiple quiz questions and their options.
3. Display questions one by one.
4. Accept an answer from the user.
5. Compare the entered answer with the stored correct answer.
6. Increase the score when the answer is correct.
7. Display the final score and percentage.
8. Display a performance message based on the final score.

The project is intentionally simple so that the focus remains on demonstrating core Python concepts and a complete input-processing-output workflow.



4 Existing and Proposed Solution

Existing approach

A basic quiz can be conducted manually or through a static list of questions. However, a manual approach does not automatically evaluate responses or calculate a final score. More advanced quiz platforms may provide graphical interfaces, databases, user accounts and analytics, but these are beyond the scope of this introductory Python project.

Proposed solution

The proposed solution is a lightweight Python-based command-line Quiz Game. Questions are represented using Python dictionaries and stored in a list. A loop processes each question, prints its options, accepts the user's answer and compares it with the predefined correct answer.

Value addition

The project converts a simple set of questions into an interactive application. It provides instant feedback and automatic score calculation while remaining easy to understand and modify. New questions can be added by extending the question list.

4.1 Code Submission (GitHub)

The project code and notebook are maintained in the GitHub repository:

<https://github.com/rajoshi123/upskillcampus>

4.2 Report Submission

The final report can be added to the same repository after submission as part of the internship documentation.



5 Proposed Design / Model

The application follows a straightforward sequence: initialize the quiz data and score, display the welcome message, iterate through the questions, accept and evaluate the user's answer, update the score, and finally calculate the percentage and display the performance result.

5.1 High Level Diagram

START → Initialize score → Load questions → Display question and options → Accept answer → Compare answer → Update score → Repeat for remaining questions → Calculate percentage → Display result → END

5.2 Low Level Design

Each question is represented as a dictionary containing three main fields: question text, a list of options, and the correct answer. All question dictionaries are stored in a list. The program accesses these fields during each iteration.

5.3 Interfaces / Flow

The user interface is text based. The primary interaction is through keyboard input using the answer choices A, B, C or D. The output consists of the question, available choices, correctness feedback and the final score. This keeps the interface simple and suitable for a beginner Python project.



6 Performance Test

Performance testing for this project focuses primarily on functional correctness, input handling and predictable output rather than high-performance computing metrics. Since the application is a small local command-line program, execution time and memory requirements are minimal for the intended use.

6.1 Test Plan / Test Cases

TC01 — Start the program and verify that the welcome message appears.

TC02 — Enter a correct answer and verify that the score increases.

TC03 — Enter an incorrect answer and verify that the score does not increase.

TC04 — Complete all questions and verify that the final score equals the number of correct answers.

TC05 — Verify percentage and performance message for the completed quiz.

6.2 Test Procedure

The program was executed in the Python/Jupyter environment. Each question was displayed sequentially. Answers were entered using the required A/B/C/D format. The program displayed correctness feedback and continued to the next question. After the final question, the score, percentage and performance message were checked.

6.3 Performance Outcome

A complete five-question test run produced a score of 5/5 and a percentage of 100.0%. The program correctly displayed the completion message and the performance message 'Excellent performance!'. This confirms that the main quiz flow, answer checking, score calculation and final result display were functioning as expected for the tested run.



7 My Learnings

The internship strengthened my understanding of Python fundamentals and showed me how individual programming concepts combine to create a working application.

Technical learnings

- Python variables, strings, lists and dictionaries.
- Conditional statements and comparison logic.
- Loops for repeated processing.
- User input and formatted output.
- Score calculation and percentage calculation.
- Organizing questions as structured data.
- Running and testing Python programs in Jupyter Notebook.
- Maintaining both notebook and Python source-code versions.
- Using GitHub as a repository for project submission.

Problem-solving learnings

I learned to divide a project into smaller steps instead of attempting the complete application at once. I also became more comfortable identifying errors from program output and checking the logic of individual components.

Career relevance

The internship gave me practical exposure to the development cycle of a small software project. The experience will help me approach future projects with greater confidence, especially when working with Python, data processing, automation and other programming-based applications.



8 Future Work Scope

The current Quiz Game is a basic command-line implementation. Several improvements can be considered as future work if the project is extended.

1. Graphical User Interface — A GUI using Tkinter or another Python framework could make the quiz more interactive and visually appealing.
2. Larger Question Bank — Questions could be loaded from external files or a database instead of being hard-coded.
3. Randomization — Questions and options could be randomized to create a different quiz sequence for each attempt.
4. Difficulty Levels — Questions could be grouped into easy, medium and difficult categories.
5. User Profiles — A future version could store user names and previous scores.
6. Persistent Scoreboard — Scores could be saved in a database or file for later comparison.
7. Timer — A time limit could be introduced for individual questions or the complete quiz.
8. Data Analytics — Attempt history could be analyzed to identify frequently missed questions and overall performance.

Conclusion

The Python Quiz Game successfully demonstrates the application of fundamental Python programming concepts in a complete, interactive program. The project met its primary requirements and was tested successfully with a five-question run producing 5/5 and 100%.

The internship experience helped me move from learning individual Python concepts to applying them in a practical project. It also provided exposure to Jupyter Notebook, source-code management and GitHub submission. The knowledge and problem-solving skills gained through this work provide a useful foundation for developing more advanced Python applications in the future.